data200-midterm-project

March 3, 2026

1 Predicting Real Estate Value: Analysis of House Features vs. Market Trends

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- Link to Project on Github: GithubRepo_Group20

[ ]:
```
import statsmodels.api as sm
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import re
import statsmodels
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

ames_df = pd.read_csv('train.csv')
hpi_df = pd.read_csv('ATNHPIUS11180Q.csv')
```

1.1 Introduction

When looking to buy a home, people naturally look at its physical features: square footage, age, and building quality. However, a house does not exist in a vacuum, with its value also being influenced by the broader house economy in the area.

Research Question: How well can house prices be predicted using property-specific features and macroeconomic data? Which features micro (property details) or macro (House Price Index) trends contribute most to price variation?

To answer this, we will merge physical housing data from Ames, Iowa, with economic index data from the Federal Reserve, visualize the historical trends, and fit a model to definitively find exactly how much each feature is "worth" in dollars.

2 Data Sources Used

The Micro-Data (Ames Housing Data) Contains physical characteristics and final sale price of houses in Ames.

The Macro-Data (Housing Price Index) Housing Price data from the Federal Reserve. Gives quarterly score that shows the overall health of the housing market.

3 Code: Setup and Joining

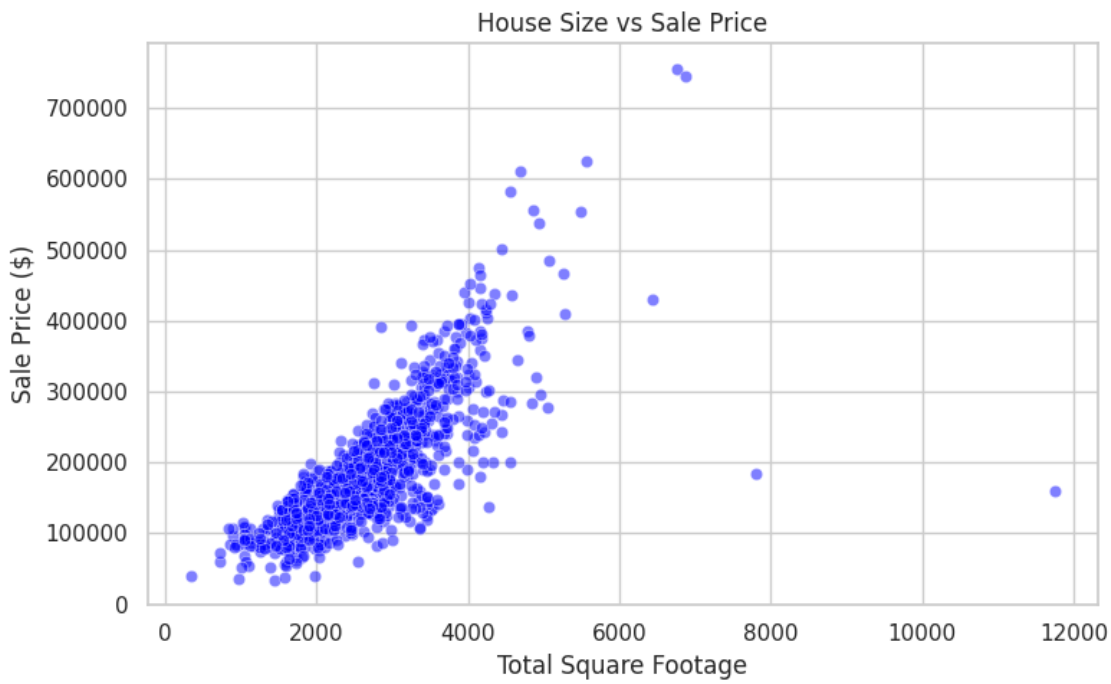
The house sales data set contains data of what day the houses were sold, while the house price index is reported quarterly, so we are going to group the months into financial quarters. We are then going to left join the two datasets, assigning a specific economic "score" to every single house based on the quarter it was sold in.

[ ]:
```
#Combine all floors and basement into Total Square Footage
ames_df["Total_Square_Footage"] =␣
↪ames_df["TotalBsmtSF"]+ames_df["1stFlrSF"]+ames_df["2ndFlrSF"]

#Convert month sold into a financial quarter
def get_quarter_start(month):
  if month in [1,2,3]: return "01" # Quarter 1
  if month in [4,5,6]: return "04" # Quarter 2
  if month in [7,8,9]: return "07" # Quarter 3
  if month in [10,11,12]: return "10" # Quarter 4

ames_df["QuarterMonth"] = ames_df["MoSold"].apply(get_quarter_start)

#Make a stadard date string (YYYY-MM-DD) and convert both to date time

ames_df["Quarter_start"] = ames_df["YrSold"].astype(str) + "-" +␣
↪ames_df["QuarterMonth"] + "-01"
ames_df["Quarter_start"] = pd.to_datetime(ames_df["Quarter_start"])
hpi_df["observation_date"] = pd.to_datetime(hpi_df["observation_date"])

#Join both datasets

augmented_df = pd.merge(left = ames_df, right = hpi_df, left_on =␣
↪"Quarter_start", right_on = "observation_date", how = "left")

augmented_df.rename(columns= {"ATNHPIUS11180Q":"HPI_Score"}, inplace = True)
```

4 Exploratory Data Analysis - Visualizations

4.1 Visual 1: House Size vs. Sale Price

[ ]:
```
sns.set_theme(style = "whitegrid")
plt.figure(figsize = (8,5))
sns.scatterplot(data = augmented_df, x = "Total_Square_Footage", y =␣
 ↪"SalePrice", alpha = 0.5, color = "blue")
plt.title("House Size vs Sale Price")
plt.xlabel("Total Square Footage")
plt.ylabel("Sale Price ($)")
plt.tight_layout()
plt.savefig("lr_size_vs_price.png")
plt.show()

correlation = ames_df["SalePrice"].corr(ames_df["Total_Square_Footage"])
print("The correlation is:", correlation)
```




```
The correlation is: 0.7822600527979845
```

Interpretation (Plot 1)

This scatterplot shows that there is a strong positive correlation of 0.7823 between the total square footage of the respective house, and its sale price. Meaning that as the sale price of the house goes up, the total square footage of the house also goes up, and vice-versa. Of course, we can see that there are some outlier cases.

4.2 Visual 2: Overall Quality vs Sale Price

Since Overall Quality is a categorical variable, we decided to use a boxplot to help visualize the distribution of sale prices accross each quality level.

[ ]:
```
sns.set_theme(style="whitegrid")
plt.figure(figsize=(12, 6))

sns.boxplot(x='OverallQual', y='SalePrice', data=augmented_df,␣
 ↪palette='Spectral')

plt.title('Impact of Overall Quality on Sale Price', fontsize=15)
plt.xlabel('Overall Quality (1: Very Poor, 10: Very Excellent)', fontsize=12)
plt.ylabel('Sale Price ($)', fontsize=12)
plt.tight_layout()

plt.savefig("overall_qual_boxplot.png")
plt.show()
```

```
/tmp/ipykernel_177/3964866919.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.boxplot(x='OverallQual', y='SalePrice', data=augmented_df,
palette='Spectral')
```

Interpretation (Plot 2)

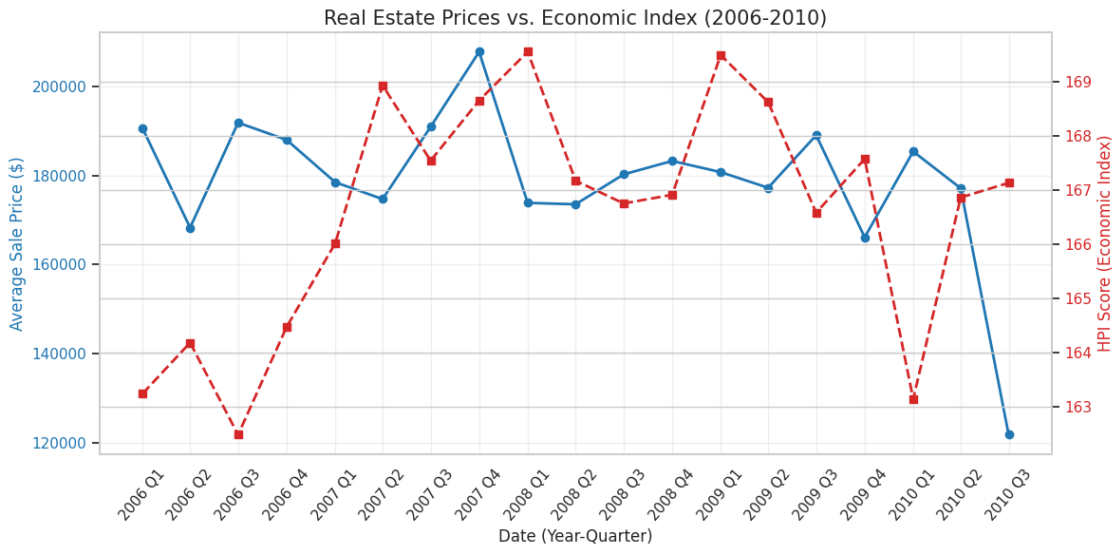
Since we have a continuous upward trend when increasing from one quality level to the next, we

can conclude that there is a strong positive correlation between the overall quality of the house and the sale price. The correlation is not strictly linear as there is a much bigger jump between the sale price and overall quality at the higher quality level than the lower ones. This suggest that premium finishes on a house command a much higher market premium. Aditionally, the boxes get larger as the quality of the houses get greater indicating that there is a greater volatility on the sale price of homes indicating that micro features such as location play a more significant role in determining the market price for premium houses.

4.3 Visual 3: Real Estate Prices vs. Economic Index

[ ]:
```
trend_df = augmented_df.groupby("Quarter_start").agg({
    "SalePrice": "mean",
    "HPI_Score": "mean"
}).reset_index()

# fig = drawing paper, ax1 = the area of actual graph (x,y axis)
fig, ax1 = plt.subplots(figsize=(12, 6))


# Axis 1: Average Sale Price
color1 = 'tab:blue'
ax1.set_xlabel('Date (Year-Quarter)', fontsize=12)
ax1.set_ylabel('Average Sale Price ($)', color=color1, fontsize=12)
ax1.plot(trend_df['Quarter_start'], trend_df['SalePrice'], color=color1,␣
 ↪marker='o', linewidth=2, label='Avg Sale Price')
trend_df["Quarter_label"] = (
    trend_df["Quarter_start"].dt.year.astype(str) + " Q" +
    trend_df["Quarter_start"].dt.quarter.astype(str)
)
ax1.set_xticks(trend_df["Quarter_start"])
ax1.set_xticklabels(trend_df["Quarter_label"], rotation=50)
ax1.tick_params(axis='y', labelcolor=color1)
ax1.grid(True, alpha=0.3)

# Axis 2: HPI Score (The Economic Index)
ax2 = ax1.twinx()
color2 = 'tab:red'
ax2.set_ylabel('HPI Score (Economic Index)', color=color2, fontsize=12)
ax2.plot(trend_df['Quarter_start'], trend_df['HPI_Score'], color=color2,␣
 ↪marker='s', linestyle='--', linewidth=2, label='HPI Score')
ax2.tick_params(axis='y', labelcolor=color2)
plt.xticks(rotation=45)

# Title and Layout
plt.title('Real Estate Prices vs. Economic Index (2006-2010)', fontsize=15)
fig.tight_layout()
```

```
plt.savefig("price_vs_hpi_trend.png")
plt.show()
# Showing trend: looks like Avg Sale price, and HPI have opposite trends
```



Interpretation (Plot 3)

The dual axis graph above shows a visible inverse relationship between average sale prices (blue line), and national House Price Index (red line) between 2006 and 2010. While the macro-level HPI score shows an upward climb peaking in early 2008, average sale prices trended in the opposite direction during the same periods. This negative correlation is also shown in 2007, and also in 2010 when the Economic Index had massive volatility before a crash in early 2010(Q1) while the Ames average prices hit its ultimate five-year low later that year in 2010(Q3). Ultimately, these diverging trends suggest that micro-level property details and localmarket dynamics in Ames played a more crucial role in price changes compared to broader national macroeconomic trends during this period.

5 Model Plots

5.0.1 Micro Model (OLS)

Model 1: where it shows the correlation between Micro variables and Sale Prices. Model 1 serves as a baseline regression including only micro-level property characteristics to measure their explanatory power on housing prices.

Imagine you want to guess the price of a house. You know it starts at a certain base value, and then you add or subtract money for every specific feature. The OLS model tries to find the perfect "price tag" for each of those features. It creates an equation that looks like this:

Estimated House Price = (Base Value) + (Value of 1 Sq Ft × Total Square Feet) + (Value of Quality Rating × Overall Quality) + (Value of a Garage × Number of Garage Cars) …

The OLS method looks at all the historical house sales in the dataset and calculates those "price tags" (coefficients) by minimizing the errors. It tweaks the values until its estimated prices are as close to the actual sale prices as mathematically possible.

[ ]:
```
# Model 1 (Micro model)

X_micro = augmented_df[["GrLivArea", "OverallQual", "TotalBsmtSF",␣
 ↪"GarageCars", "YearBuilt", "FullBath"]]
X_micro = sm.add_constant(X_micro)
y = augmented_df["SalePrice"]

model_micro = sm.OLS(y, X_micro).fit()
print(model_micro.summary())\

#Extract model and put it into a percentage
variance_explained_pct = model_micro.rsquared * 100

print("\n-------------------------------------------")
print(f"Variance Explained (R-squared): {variance_explained_pct:.4f}%")
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              SalePrice   R-squared:                       0.769
Model:                            OLS   Adj. R-squared:                  0.768
Method:                 Least Squares   F-statistic:                     804.6
Date:                Mon, 02 Mar 2026   Prob (F-statistic):               0.00
Time:                        18:44:28   Log-Likelihood:                -17475.
No. Observations:                1460   AIC:                         3.496e+04
Df Residuals:                    1453   BIC:                         3.500e+04
Df Model:                           6
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -7.377e+05   9.05e+04     -8.154      0.000   -9.15e+05    -5.6e+05
GrLivArea     54.0758      3.000     18.024      0.000      48.191      59.961
OverallQual 2.052e+04   1156.785     17.739      0.000    1.83e+04    2.28e+04
TotalBsmtSF   29.1821      2.843     10.263      0.000      23.605      34.760
GarageCars  1.465e+04   1822.657      8.036      0.000    1.11e+04    1.82e+04
YearBuilt    336.5467     47.643      7.064      0.000     243.091     430.002
FullBath   -5503.9737   2650.906     -2.076      0.038   -1.07e+04    -303.962
==============================================================================
Omnibus:                      458.824   Durbin-Watson:                   1.979
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            49948.644
Skew:                          -0.372   Prob(JB):                         0.00
Kurtosis:                      31.645   Cond. No.                     2.47e+05
==============================================================================
```

```
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The condition number is large, 2.47e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

--------------------------------------------
Variance Explained (R-squared): 76.8647%
```

The model achieves an R^2 (how much of the puzzle is successfully solved) of 0.769, meaning that approximately 76.9% of the differences in house prices can be explained by the specific micro-level variables we included (like living area, overall quality, year built, garage capacity, etc.). Furthermore, all variables in this model are statistically significant (p-values < 0.05), indicating strong evidence that they meaningfully contribute to price determination.

Interpretation of the Micro Model:

- These features have a strong positive impact on valuation. Holding other variables constant, every additional square foot of living area (GrLivArea) increases the house price by approximately $54.
- Similarly, a one-unit increase in overall material and finish quality (OverallQual) adds about $20,520 to the home's value.
- Interestingly, adding a full bathroom (FullBath) decreases the estimated house price by $5,503. Because this model holds the total living area constant, this negative coefficient suggests that squeezing in an extra bathroom removes valuable square footage from other primary living spaces, which negatively impacts the home's overall value.

5.0.2 Macro Model (Model 2)

[ ]:
```
augmented_df["SalePrice"].corr(augmented_df["HPI_Score"])
```

[ ]:
```
np.float64(-0.02234874590159313)
```

[ ]:
```
# # Model 2 (Macro model)
X_macro = augmented_df[["HPI_Score"]]
X_macro = sm.add_constant(X_macro)
y = augmented_df["SalePrice"]

model_macro = sm.OLS(y, X_macro).fit()
#print(model_macro.summary())

# Extract R-squared and turn it into a percentage
r_squared_pct = model_macro.rsquared * 100

print(f"Percentage of variance explained: {r_squared_pct:.4f}%")
```

```
Percentage of variance explained: 0.0499%
```

Interpretation of Model 2: This model shows that the national House Price Index (HPI) explains only a 0.05% of the reasons why house prices change in Ames, which is extremely small. Therefore, the overall national economy is almost useless in reality for predicting the price of a house.

5.0.3 Standardized Model:

In the first model, it is impossible to compare the variables directly to see which one is the "most important." GrLivArea is measured in square feet, GarageCars is measured in whole cars, and OverallQual is on a 1-10 scale. Standardizing the data converts everything into the same universal unit: standard deviations, which will help us find our most valuable predictor.

[ ]:
```
# 1. Define X using your original variables (without the constant)
X = augmented_df[["GrLivArea", "OverallQual", "TotalBsmtSF", "GarageCars",␣
 ↪"YearBuilt", "FullBath"]]
y = augmented_df["SalePrice"]

# 2. Standardize both X and y
X_std = (X - X.mean()) / X.std()
y_std = (y - y.mean()) / y.std()

# 3. Add the constant AFTER standardizing
X_std = sm.add_constant(X_std)

# 4. Run the regression
model_std = sm.OLS(y_std, X_std).fit()

print(model_std.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              SalePrice   R-squared:                       0.769
Model:                            OLS   Adj. R-squared:                  0.768
Method:                 Least Squares   F-statistic:                     804.6
Date:                Mon, 02 Mar 2026   Prob (F-statistic):               0.00
Time:                        18:44:36   Log-Likelihood:                -1002.6
No. Observations:                1460   AIC:                             2019.
Df Residuals:                    1453   BIC:                             2056.
Df Model:                           6
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const        1.353e-16      0.013   1.07e-14      1.000      -0.025       0.025
GrLivArea       0.3577      0.020     18.024      0.000       0.319       0.397
OverallQual     0.3572      0.020     17.739      0.000       0.318       0.397
TotalBsmtSF     0.1612      0.016     10.263      0.000       0.130       0.192
GarageCars      0.1378      0.017      8.036      0.000       0.104       0.171
YearBuilt       0.1280      0.018      7.064      0.000       0.092       0.163
```

```
FullBath       -0.0382      0.018     -2.076      0.038      -0.074      -0.002
==============================================================================
Omnibus:                      458.824   Durbin-Watson:                   1.979
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            49948.644
Skew:                          -0.372   Prob(JB):                         0.00
Kurtosis:                      31.645   Cond. No.                         4.02
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
```

Interpretation of Standardized Model

With an R^2 of 0.769, the model captures a substantial portion of the variation in house prices. Above-ground living area (GrLivArea) has the largest standardized coefficient (0.3577). This shows that, relative to the other features in this model, changes in above-grade living space contribute the most to price variation. More specifically, a one-standard-deviation increase in above ground living area results in a 0.3577 standard-deviation increase in the sale price. Therefore, GrLivArea is the most valuable predictor among the micro-level variables.

6 Conclusion

Returning back to our research question, "How well can house prices be predicted using property-specific features and macroeconomic data? Which features micro (property details) or macro (House Price Index) trends contribute most to price variation?"

We concluded that the house prices in the Ames dataset can be predicted highly effectively using property specific micro features, with a linear regression model achieving an R^2 of 0.769, meaning 76.9% of price variation is explained by variables like above-ground living area, basement size, year built, and full bathrooms. This is evident in Visual 3, in which the macro-level HPI score and the local average sale prices display a clear inverse relationship. Furthermore, our regression models mathematically confirmed this divergence, revealing that the broader economy explains just 0.05% of house price changes. Ultimately, our research proves that buyers prioritize tangible micro features most notably the aboveground living area far more than the macroeconomic climate.

6.1 Biases in our research

Our project is focused on just one housing markert, the city of Ames, Ohio, some potential biases that arised are:

- The variables available and how they are measured for the quality scores, which are really subjective and coded us.
- We may have overstated the importance of some features while ignoring others such as school quality and neighbourhood desirability.
- Lastly, selection bias arises because only sold properties are observed, hence not all homes in the market are observed